

How Does BI Rate Affect the Rupiah Exchange Rate? An Analysis of Bank Indonesia's Interest Rate Policy, Inflation, and Exchange Rate Movements (2017–2026)

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Abstract. Monetary policy plays a crucial role in maintaining economic stability, controlling inflation, and influencing exchange-rate movements. In Indonesia, Bank Indonesia (BI) utilizes the BI Rate as its primary monetary policy instrument to manage inflation expectations and support the stability of the Rupiah. This project aims to explore the relationship between BI Rate movements, inflation trends, and the Rupiah exchange rate from 2017 to 2026. Using historical economic data obtained from Bank Indonesia and Statistics Indonesia (BPS), the analysis employs exploratory data analysis (EDA), trend analysis, and interactive dashboard visualization developed in Tableau. The findings indicate that periods of increasing BI Rates generally coincide with efforts to control inflationary pressures and strengthen investor confidence in the Indonesian economy. However, exchange-rate movements are influenced not only by domestic monetary policy but also by external factors such as global economic conditions, capital flows, and market sentiment. The study highlights the importance of balanced monetary policy in achieving both price stability and sustainable economic growth.

Keywords - Data Cleaning, Exploratory Data Analysis, PostgreSQL, Manajemen Operasional, Perilaku Konsumen Digital.

I. PENDAHULUAN

Central banks play an essential role in maintaining economic stability through monetary policy. One of the primary responsibilities of Bank Indonesia is to maintain price stability while supporting sustainable economic growth. To achieve these objectives, Bank Indonesia utilizes the BI Rate as its primary monetary policy instrument. Changes in interest rates influence borrowing costs, consumer spending, business investment, and capital flows, making them an important tool for managing economic activity and inflation.

Interest-rate adjustments can also affect exchange-rate movements. When inflation rises above target levels, central banks often increase interest rates to reduce inflationary pressures and stabilize financial markets. Higher interest rates may attract foreign capital inflows, increasing demand for domestic currency and potentially strengthening the exchange rate. Conversely, during periods of economic slowdown, interest rates may be reduced to encourage borrowing, spending, and investment, which can support economic growth but may also create inflationary pressures. In Indonesia, the stability of the Rupiah exchange rate is closely monitored because it affects import prices, export competitiveness, investor confidence, and overall economic performance. Significant fluctuations in the Rupiah can influence domestic prices and create uncertainty within financial markets. As a result, understanding the relationship between BI Rate movements and exchange-rate dynamics is important for evaluating the effectiveness of monetary policy.

This project aims to analyze the relationship between BI Rate, inflation, and the Rupiah exchange rate using historical data from 2017 to 2026. The analysis focuses on identifying observable trends and patterns through exploratory data analysis (EDA) and interactive dashboard visualization developed in Tableau. Specifically, the study examines how BI Rate movements correspond with changes in inflation and exchange-rate behavior over time, while recognizing that exchange-rate fluctuations are influenced by a variety of domestic and global economic factors.

The analysis uses three key macroeconomic indicators: BI Rate, inflation rate, and the USD/IDR exchange rate. Rather than establishing direct causal relationships through advanced econometric techniques, this project adopts a descriptive and exploratory approach to provide insights into monetary-policy trends and their potential relationship with exchange-rate stability. Through interactive visualizations and data-driven storytelling, the project seeks to demonstrate how monetary-policy decisions may contribute to broader macroeconomic outcomes in Indonesia.

II. METODE

This project utilizes secondary data obtained from publicly available government institutions in Indonesia. The BI Rate data were collected from Bank Indonesia, while inflation data were obtained from Statistics Indonesia (BPS). Exchange-rate data, represented by the USD/IDR currency pair, were also sourced from Bank Indonesia. These institutions provide reliable and authoritative macroeconomic indicators that are widely used for economic analysis and policy evaluation.

The analysis covers the period from 2017 to 2026, allowing the study to capture multiple monetary-policy cycles and significant economic events that may influence inflation and exchange-rate movements. This time horizon provides sufficient historical context to observe trends, fluctuations, and potential relationships among the selected variables.

Three key variables are examined in this study. The first is the BI Rate, which serves as Bank Indonesia's benchmark interest rate and primary monetary-policy instrument. The second is the inflation rate, measured as the annual percentage change in consumer prices. The third is the USD/IDR exchange rate, which reflects the value of the Indonesian Rupiah relative to the United States Dollar and serves as an indicator of currency stability.

Prior to analysis, several data-preparation procedures were conducted to ensure data quality and consistency. These procedures included data collection from official sources, validation of records across datasets, consistency checks, handling of missing values when necessary, standardization of date formats, and transformation of data structures to facilitate visualization and dashboard development.

The analytical process follows a structured workflow consisting of data collection, data cleaning, exploratory data analysis (EDA), trend analysis, and dashboard development. Exploratory analysis was performed to identify patterns, trends, and anomalies within the dataset, while trend analysis focused on examining long-term movements in the BI Rate, inflation rate, and exchange rate. The final stage involved the development of an interactive Tableau dashboard designed to communicate insights effectively through data visualization and storytelling. The methodology adopted in this project is descriptive and exploratory in nature, emphasizing the interpretation of historical trends rather than establishing causal relationships through econometric modeling.

III. HASIL DAN PEMBAHASAN

The analysis reveals several notable trends in Indonesia's monetary and macroeconomic indicators during the 2017–2026 period. The BI Rate experienced multiple adjustments throughout the observation period, reflecting Bank Indonesia's efforts to maintain macroeconomic stability in response to changing domestic and global economic conditions. Periods of higher interest rates generally corresponded with inflation-control measures and initiatives aimed at supporting financial-market stability. Conversely, lower interest-rate periods were typically implemented to encourage economic activity, increase borrowing, and stimulate investment.

Inflation remained relatively stable during several years but also experienced fluctuations driven by domestic and international economic developments. Maintaining inflation within a targeted range is a key objective of monetary policy because excessive inflation can reduce purchasing power, increase uncertainty, and negatively affect economic welfare. The BI Rate serves as one of Bank Indonesia's primary instruments for influencing inflation dynamics and maintaining price stability.

The Rupiah exchange rate against the US Dollar displayed both appreciation and depreciation throughout the analysis period. Exchange-rate movements reflected a combination of factors, including domestic monetary policy, global interest-rate conditions, commodity-price fluctuations, investor sentiment, and international capital flows. These factors collectively contribute to currency volatility and influence the effectiveness of monetary-policy transmission.

A comparison of BI Rate movements and exchange-rate trends suggests that interest-rate policy plays an important role in supporting currency stability. Higher interest rates are often associated with efforts to strengthen the Rupiah by attracting foreign capital and increasing investor confidence in domestic financial assets. However, the relationship between interest rates and exchange rates is not always linear, as exchange-rate movements are influenced by numerous economic and external factors beyond monetary policy alone.

The findings can be further interpreted through standard monetary-policy transmission mechanisms. When Bank Indonesia increases the BI Rate, borrowing costs generally rise, leading to lower consumer spending and reduced inflationary pressure. Higher interest rates may also support the Rupiah by increasing the attractiveness of domestic financial assets to investors. Conversely, when the BI Rate is reduced, borrowing becomes more affordable, encouraging consumption and investment activities that can stimulate economic

growth. While lower interest rates may support economic expansion, they can also increase inflationary pressure if demand grows faster than productive capacity.

Overall, the results highlight the trade-offs faced by policymakers in balancing economic growth, inflation control, and exchange-rate stability. The analysis demonstrates that the BI Rate remains an important monetary-policy instrument, although its effectiveness is influenced by broader domestic and global economic conditions. The Tableau dashboard provides an interactive platform for exploring these relationships and understanding how monetary-policy decisions are reflected in Indonesia's macroeconomic performance over time.

IV. DASHBOARD INSIGHTS

To enhance the communication of analytical findings, an interactive Tableau dashboard was developed to visualize the relationships between BI Rate movements, inflation trends, and USD/IDR exchange-rate fluctuations. The dashboard is designed to provide a comprehensive overview of Indonesia's monetary-policy environment through intuitive visualizations and interactive features that support data exploration.

The dashboard consists of several key components, including KPI summary cards that present the latest macroeconomic indicators, trend visualizations for BI Rate, inflation, and exchange-rate movements, a monetary-policy impact infographic, and a dedicated key-takeaways section. Together, these elements enable users to examine historical patterns and better understand the potential interactions among the selected variables.

Several important insights emerged from the dashboard analysis. First, the BI Rate serves as Bank Indonesia's primary monetary-policy instrument for managing inflation expectations and maintaining overall economic stability. Changes in the benchmark interest rate reflect policy responses to evolving economic conditions and inflationary pressures.

Second, periods of higher interest rates generally coincide with efforts to stabilize inflation and support the Rupiah. By increasing borrowing costs and attracting capital inflows, higher interest rates may contribute to stronger currency performance and improved confidence in domestic financial markets.

Third, exchange-rate movements are influenced by a combination of domestic monetary policy and external economic factors. While interest-rate adjustments can affect investor behavior and capital flows, global financial conditions, commodity prices, and international economic developments also play significant roles in determining currency performance.

Finally, the dashboard highlights the complex trade-offs involved in monetary-policy decision-making. Policymakers must continuously balance the objectives of controlling inflation, maintaining exchange-rate stability, and supporting sustainable economic growth. The interactive dashboard provides a practical tool for exploring these relationships and translating macroeconomic data into accessible and actionable insights.

V. CONCLUSION

This project examined the relationship between BI Rate movements, inflation trends, and Rupiah exchange-rate fluctuations between 2017 and 2026.

The analysis demonstrates that BI Rate plays an important role in Indonesia's monetary-policy framework and contributes to efforts aimed at maintaining macroeconomic stability. While higher interest rates are often associated

with inflation control and exchange-rate support, exchange-rate dynamics remain affected by broader global economic conditions.

Overall, the findings suggest that BI Rate should be viewed as an important—but not exclusive—determinant of exchange-rate movements. Effective monetary policy requires balancing inflation control, financial-market stability, and long-term economic growth.

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